



UNIVERZITET U NOVOM SADU
FAKULTET TEHNIČKIH NAUKA
KATEDRA ZA AUTOMATIKU I UPRAVLJANJE SISTEMIMA

Termički sistemi

Modeli fizičkih sistema

Modeliranje i simulacija sistema

Termički sistemi

- su sistemi gde postoji skladištenje ili prenos toplote
- Matematički modeli se izvode na osnovu poznatih zakona termodinamike
- Primeri: termometar, sistem za rashlađivanje, pećnice, frižideri, ...
- Matematički model čine
 - zbog prostorne distribuiranosti →
parcijalne diferencijalne jednačine
 - koncentrisani parametri →
obične dif. jednačine
 - linearizacija → linearne obične dif. jednačine
- Posmatramo sisteme
 - Bez promene agregatnih stanja
 - ne postoje termodinamički ciklusi (u hemijskim procesima)

aproksimacija

aproksimacija

Promenljive

- Temperatura – θ [K]
 - obično je u različitim tačkama tela različita
 - smatraćemo da je θ u svim tačkama tela ista i jednaka posečnoj temperaturi tela
 - ukoliko je bitno da pojedini delovi tela imaju različite θ , onda se telo može posmatrati iz više segmenata
 - obično se θ bira za promenljive stanja
 - Temperatura ambijenta (spoljašnja temp.) θ_a
- Količina toplote – q [J/s]≡[W]

Stanje ravnoteže

- stanje ravnoteže
 - gde se odvijaju normalne operacije
- devijacije promenljivih u odnosu na nominalne vrednosti su od interesa kada se posmatraju dinamičke karakteristike

$$\theta(t) = \bar{\theta} + \hat{\theta}(t)$$

$$q(t) = \bar{q} + \hat{q}(t)$$

$$\hat{\theta}(t) = \theta(t) - \bar{\theta}$$

$$\hat{q}(t) = q(t) - \bar{q}$$

Elementi u termičkom sistemu

- Dva tipa pasivnih elemenata:
 - Termička kapacitivnost
 - Termička otpornost
- Aktivan element
 - Termički izvor

Termička kapacitivnost

- Postoji algebarska zavisnost između
 - temperature tela θ i
 - akumulirane toplote u njemu Δq
- Ako nema promene agregatnog stanja i θ se menja u uskim granicama zavisnost se može posmatrati kao linearna

$$\dot{\theta}(t) = \frac{1}{C} (q_{\text{in}}(t) - q_{\text{out}}(t))$$

$$\theta(t) = \theta(t_0) + \int_{t_0}^t \frac{1}{C} (q_{\text{in}}(t) - q_{\text{out}}(t)) dt$$

- C - toplotni kapacitet tela [J/K]

$$C = m \cdot \sigma$$

$\sigma \equiv$ specifična toplota tela

- smatra se da θ tela uniformna

Termička otpornost

- Toplota se prenosi na 3 načina: provođenjem, strujanjem ili zračenjem.
- Ovde posmatramo samo provođenje toplote.

- Provođenje toplote sa jednog tela na drugo telo je srazmerno razlici temperatura dva tela

$$q(t) = \frac{1}{R} [\theta_1(t) - \theta_2(t)] \quad q(t) = \frac{1}{R} \cdot \Delta\theta$$

- $R \equiv$ termička otpornost [Ks/J]

- Karakteristika materijala

- d – debljina
- A – površina poprečnog preseka
- α – termička provodljivost (kataloški podatak)

$$R = \frac{d}{A\alpha}$$

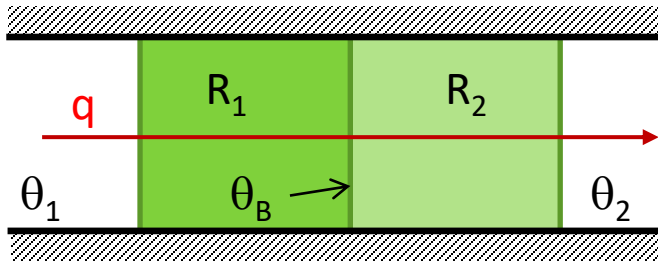
- Može se upotrebiti samo ako materijal (telo) ne akumulira energiju. Ukoliko to nije slučaj u model treba dodati i termičku kapacitivnost.

Osobine materijala

Materijal	Gustina ρ [kg/m ³]	Termička provodljivost α [W/(m·K)]	Specifična toplota tela c_p [kJ/(kg·K)]
Beton	2600	1,44-1,68	0,8
Beton armirani	2200	1,55	0,84
Gipsana ploča	600-1200	0,291-0,581	1,089
Opeka, suva	1600-1800	0,38-0,52	0,84
Opeka izolaciona	550	0,1395	-
Sneg sveže napadao	200	0,1	2,09
Staklo	2400-3200	0,582-1,047	0,77
Staklena vuna	50	0,037	0,67
Stiropor	32	0,027	1,382
Šperploča	600	0,15	2,51
Drvo bor T	546	0,16	2,7
Drvo bor -	551	0,35	2,7
Drvo hrast T	825	0,21	2,4
Drvo hrast -	890	0,36	2,4
Drvo jela T	546	0,14-0,16	2,72
Drvo jela -	-	0,35-0,72	2,72
Vazduh 0°C	1,293	0,0244	1,005
Vazduh 10°C	1,247	0,0251	1,005
Vazduh 20°C	1,205	0,0259	1,005
Vazduh 30°C	1,165	0,0267	1,005
Vazduh 40°C	1,128	0,0276	1,005

Primer

- Odrediti termičku otpornost



savršena termička izolacija

$$q = \frac{1}{R_1} [\theta_1 - \theta_B]$$

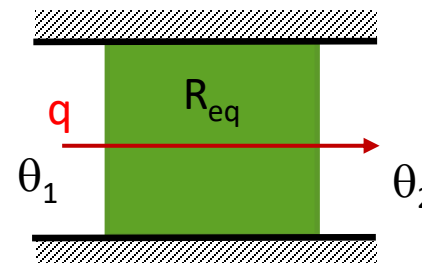
$$q = \frac{1}{R_2} [\theta_B - \theta_2]$$

$\Downarrow$

$$R_2 q + \theta_2 = \theta_B$$

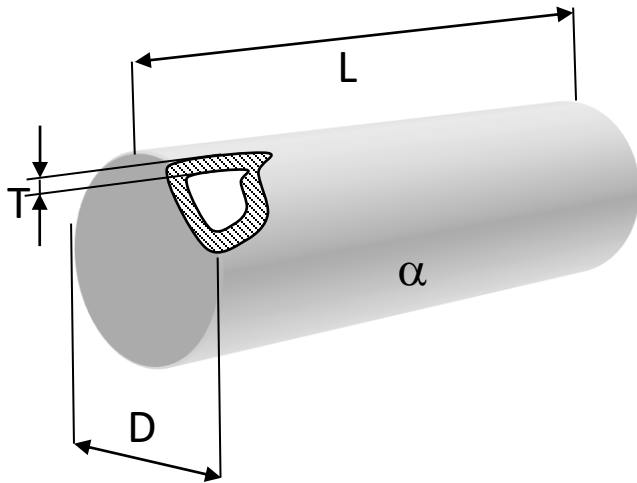
$$R_1 q = \theta_1 - \theta_2 - R_2 q \Rightarrow q = \frac{1}{R_1 + R_2} (\theta_1 - \theta_2)$$

$$R_{eq} = R_1 + R_2 \leftarrow \text{serijska veza}$$



Primer

- Odrediti termičku otpornost cilindrične posude debljine zida T i termičke provodnosti α



Term. otp. baze: $R_e = \frac{d}{A\alpha} = \frac{4T}{\pi D^2 \alpha}$

Term. otp. omotača: $R_c = \frac{T}{\pi D L \alpha}$

$$q_e = \frac{1}{R_e} \Delta\theta \quad q_c = \frac{1}{R_c} \Delta\theta$$

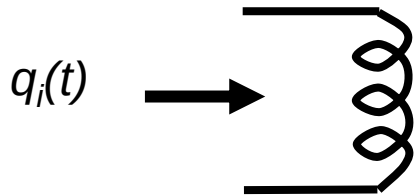
$$q_{\text{total}} = 2q_e + q_c = \left(\frac{2}{R_e} + \frac{1}{R_c} \right) \Delta\theta = \frac{1}{R_{\text{eq}}} \cdot \Delta\theta$$

$$\frac{1}{R_{\text{eq}}} = \left(\frac{1}{R_c} + \frac{2}{R_e} \right) \Rightarrow R_{\text{eq}} = \frac{R_c \cdot R_e}{2R_c + R_e}$$

Paralelna veza

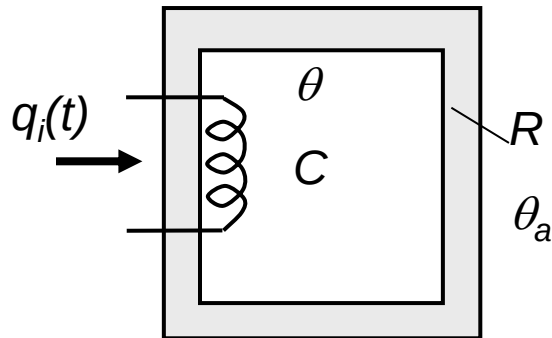
Termički izvor

- Tipovi termičkog izvora
 - izvor koji dovodi toplotu (određenom brzinom)
 - količina toplote koja se dovede je pozitivna
 - izvor koji odvodi toplotu
 - količina toplote koja se dovede je negativna
- Idealan termički izvor



Dinamički model termičkih sistema

- kao promenljive stanja se uzimaju temperature svakog tela koja ima toplotni kapacitet
- prenošenje toplote u telo sa toplotnim kapacitetom zavisi od izvora toplote i prenošenja toplote preko termičkih otpornosti.
- Primer



$$q_{\text{in}} = q_i(t)$$

$$q_{\text{out}} = \frac{1}{R} (\theta(t) - \theta_a)$$

$$\dot{\theta}(t) = \frac{1}{C} \left(q_{\text{in}}(t) - \frac{1}{R} (\theta - \theta_a) \right)$$

$$\dot{\theta}(t) + \frac{1}{RC} \theta = \frac{1}{C} q_i(t) + \frac{1}{RC} \theta_a$$

Primer (nastavak)

- Linearizacija modela

$$\dot{\theta}(t) + \frac{1}{RC} \bar{\theta} = \frac{1}{C} \bar{q}_i(t) + \frac{1}{RC} \theta_a$$

- U ustaljenom stanju:

$$\bar{\theta} = \theta_a + R\bar{q}_i$$

$$\hat{\theta}(t) = \theta(t) - \bar{\theta}$$

$$\hat{q}_i(t) = q_i(t) - \bar{q}_i$$

- Uvođenjem smena se dobija

$$\dot{\hat{\theta}}(t) + \frac{1}{RC} (\hat{\theta} + \bar{\theta}) = \frac{1}{C} [\hat{q}_i(t) + \bar{q}_i] + \frac{1}{RC} \theta_a$$

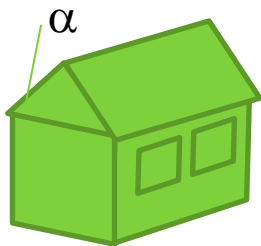
- Eliminisanja konstantnih članova na osnovu

$$\bar{\theta} = \theta_a + R\bar{q}_i$$

$$\dot{\hat{\theta}} + \frac{1}{RC} \hat{\theta} = \frac{1}{C} \hat{q}_i(t)$$

Temperatura u kući - program

```
sir = 5;  
duz = 10;  
vis = 3;  
alfa = 30/180*pi;  
brProz = 6;  
Pproz = 1;  
debKrov = 0.2;  
debZid = 0.2;  
debStaklo = 0.01;  
tpKrov = 0.037;  
tpZid = 0.037;  
tpStaklo = 0.8;  
tpVazduh = 0.025;  
sttVazduh = 1005;  
ro = 1.225;
```



```
UkPovProz = brProz * Pproz  
Pzid = 2*(sir+duz)*vis + 2*sir/2*tan(alfa)*sir/2 - UkPovProz  
Pkrov = 2*duz*sir/2/cos(alfa)  
  
V = sir * duz * vis + duz * sir/2 * sir/2*tan(alfa)  
M = ro * V  
C = M*sttVazduh  
  
Rkrov = debKrov / Pkrov / tpKrov  
Rzid = debZid / Pzid / tpZid  
Rstaklo = debStaklo / UkPovProz / tpStaklo  
Rprozor = 2*Rstaklo + 0.05 / UkPovProz / tpVazduh  
R = 1/(1/Rkrov + 1/Rzid + 1/Rprozor)  
  
Tun = 20;  
P = 0;  
[t,T] = ode45(@f, [0 3600*24], Tun, [], R,C,P);  
plot(t/3600,T), title('Temp u kuci - bez grejanja'), xlabel('sati')  
  
P = 500;  
[t,T] = ode45(@f, [0 3600*24], Tun, [], R,C,P);  
plot(t/3600,T), title('Temp u kuci - sa grejanjem'), xlabel('sati')  
  
function Tprim = f(t,T, R, C, P)  
Tspolja = 10;  
Tprim = 1/C*(-1/R*(T-Tspolja) + P);
```

Temperatura u kući – izračunati parametri

UkPovProz =
6

Pzid =
91.2169

Pkrov =
57.7350

V =
186.0844

M =
227.9534

C =
2.2909e+005

Rkrov =
0.0936

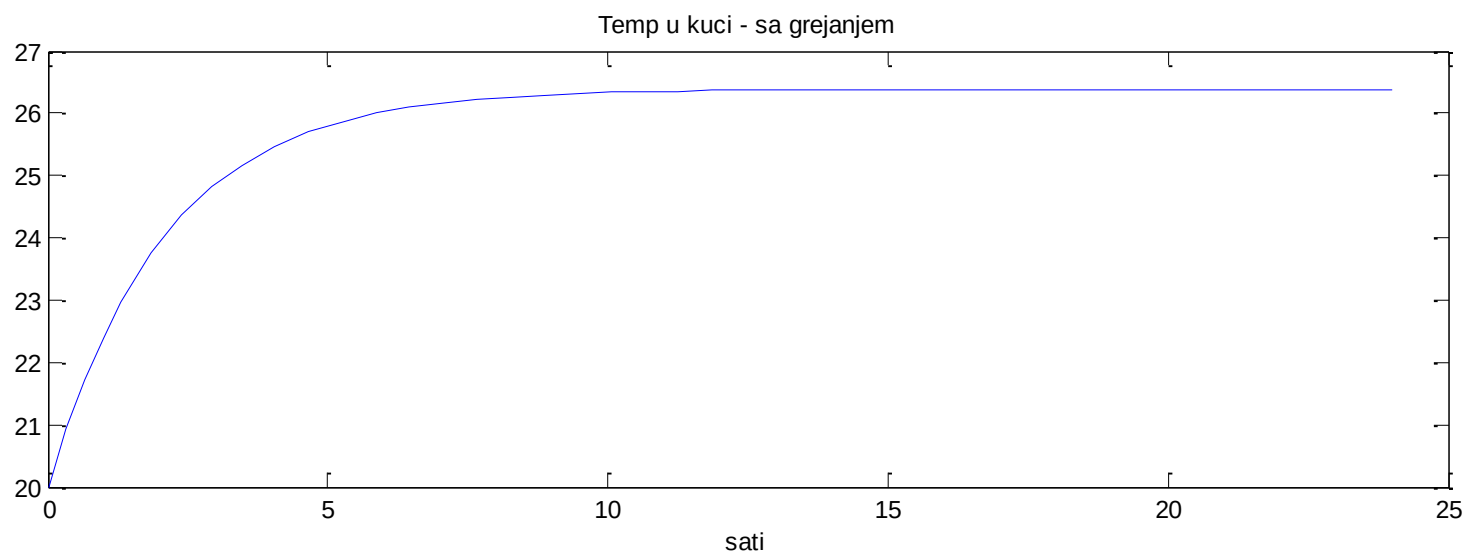
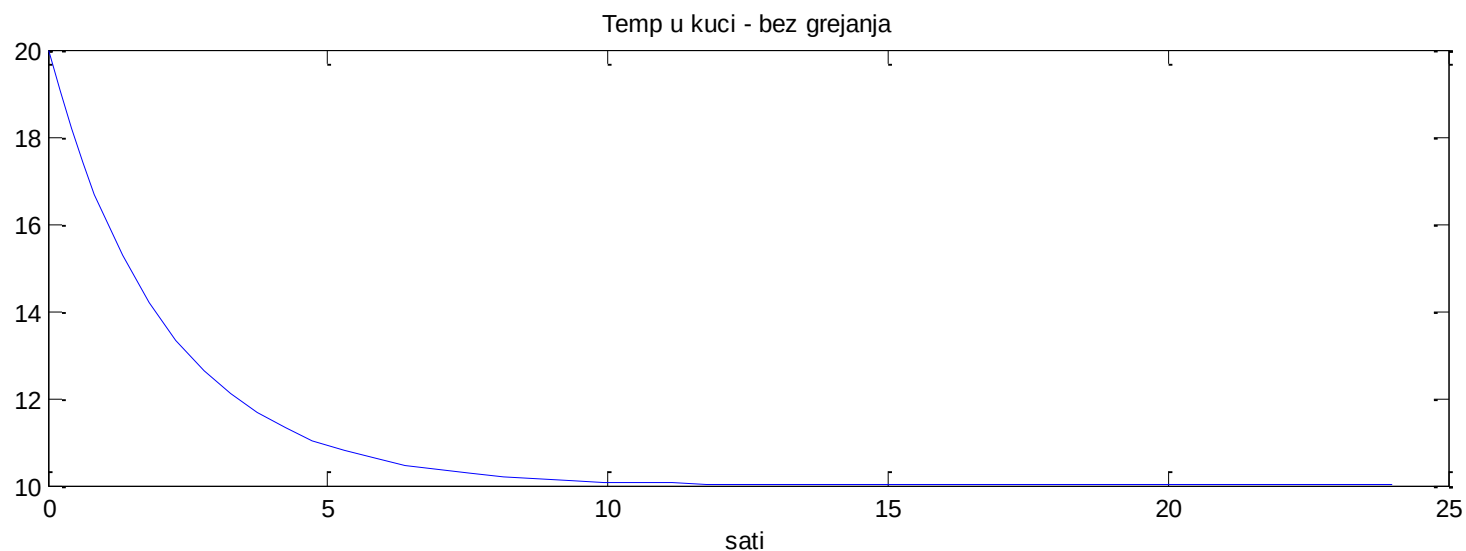
Rzid =
0.0593

Rstaklo =
0.0021

Rprozor =
0.3375

R =
0.0328

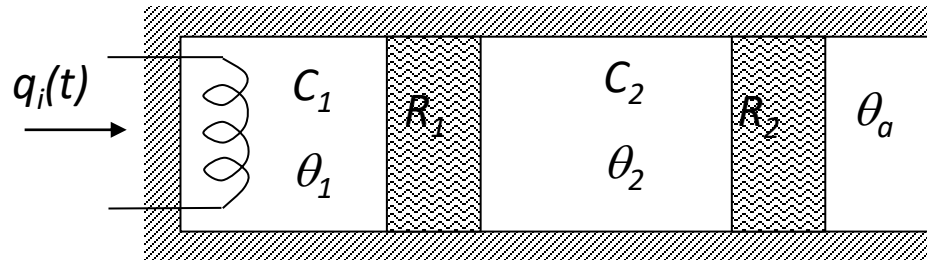
Temperatura u kući – rezultati simulacije



Primer

- Odrediti

$$\hat{\theta}_2 = f(\hat{q}_i(t)) = ?$$



$$\dot{\theta}_1(t) = \frac{1}{C_1} \left(q_i(t) - \frac{1}{R_1} (\theta_1 - \theta_2) \right) \quad \dot{\theta}_2(t) = \frac{1}{C_2} \left(\frac{1}{R_1} (\theta_1 - \theta_2) - \frac{1}{R_2} (\theta_2 - \theta_a) \right)$$

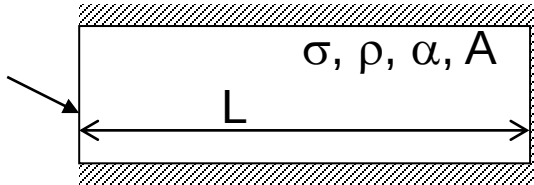
$$\left. \begin{aligned} \bar{q}_i - \frac{1}{R_1} (\bar{\theta}_1 - \bar{\theta}_2) &= 0 \\ \frac{1}{R_1} (\bar{\theta}_1 - \bar{\theta}_2) &= \frac{1}{R_2} (\bar{\theta}_2 - \theta_a) \end{aligned} \right\} \begin{aligned} \bar{\theta}_2 &= \theta_a + R_2 \bar{q}_i \\ \bar{\theta}_1 &= \theta_a + (R_1 + R_2) \bar{q}_i \end{aligned} \quad \begin{aligned} \hat{\theta}_1(t) &= \theta_1(t) - \bar{\theta}_1 \\ \hat{\theta}_2(t) &= \theta_2(t) - \bar{\theta}_2 \end{aligned}$$

$$\dot{\hat{\theta}}_1 + \frac{1}{R_1 C_1} \hat{\theta}_1 = \frac{1}{R_1 C_1} \hat{\theta}_2 + \frac{1}{C_1} \hat{q}_i$$

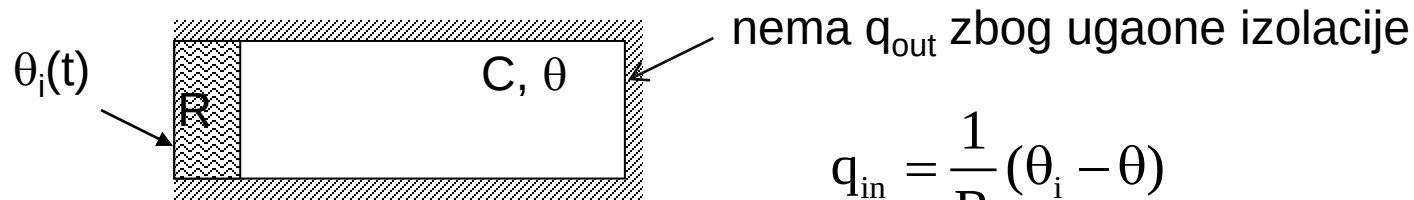
$$\dot{\hat{\theta}}_2 + \left(\frac{1}{R_1 C_1} + \frac{1}{R_2 C_2} \right) \cdot \hat{\theta}_2 = \frac{1}{R_2 C_2} \hat{\theta}_1$$

Primer

- Formirati model tela $\theta_i(t)$



- Aproksimacija 1: jedna komora



$$R = \frac{L}{A\alpha}$$

$$C = \sigma m = \sigma \rho AL$$

σ - specifična toplota tela

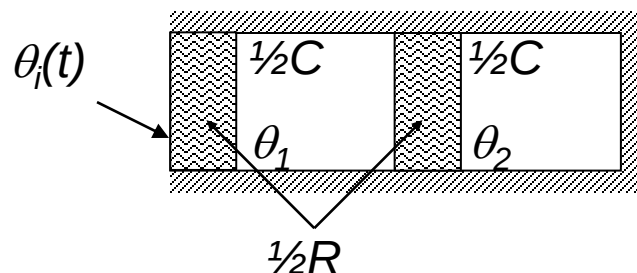
$$q_{in} = \frac{1}{R} (\theta_i - \theta)$$

$$\dot{\theta} = \frac{1}{RC} (\theta_i - \theta)$$

$$\dot{\theta} + \frac{1}{RC} \theta = \frac{1}{RC} \theta_i(t)$$

Primer (nastavak)

- Aproksimacija 2: dve komore



- Leva komora

$$q_{\text{in}} = \frac{1}{0,5R} (\theta_i(t) - \theta_1)$$

$$\dot{\theta}_1(t) = \frac{4}{RC} (\theta_i - 2\theta_1 + \theta_2)$$

- Desna komora

$$q_{\text{in}} = \frac{1}{0,5R} (\theta_1 - \theta_2) \quad q_{\text{out}} = 0$$

$$\dot{\theta}_2 = \frac{4}{RC} (\theta_1 - \theta_2)$$

$$\dot{\hat{\theta}}_1 + \frac{8}{RC} \hat{\theta}_1 = \frac{4}{RC} \hat{\theta}_2 + \frac{4}{RC} \hat{\theta}_i(t)$$

$$\dot{\hat{\theta}}_2 + \frac{4}{RC} \hat{\theta}_2 = \frac{4}{RC} \hat{\theta}_1$$

Primer (nastavak)

- Aproksimacija 3: n komora

$$\dot{\theta}_k(t) = \frac{n}{C}(q_{k+} - q_{k-}), \quad k = 1, 2, \dots, n$$

$$q_{k+} = \frac{n}{R}(\theta_{k-1}(t) - \theta_k(t)), \quad q_{k-} = \frac{n}{R}(\theta_k(t) - \theta_{k+1}(t))$$

$$\dot{\theta}_k(t) = \frac{n^2}{RC}(\theta_{k-1}(t) - 2\theta_k(t) + \theta_{k+1}(t))$$

$$q_{1+} = \frac{2}{R}(\theta_1(t) - \theta_1(t)), \quad q_{1-} = q_{2+}, \quad \dot{\theta}_1(t) = \frac{n^2}{RC}(\theta_1(t) - 2\theta_1(t) + \theta_2(t))$$

$$q_{2+} = \frac{2}{R}(\theta_1(t) - \theta_2(t)), \quad q_{2-} = q_{3+}, \quad \dot{\theta}_2(t) = \frac{n^2}{RC}(\theta_1(t) - 2\theta_2(t) + \theta_3(t))$$

...

...

...

$$q_{n+} = \frac{2}{R}(\theta_{n-1}(t) - \theta_n(t)), \quad q_{n-} = 0, \quad \dot{\theta}_n(t) = \frac{n^2}{RC}(\theta_{n-1}(t) - \theta_n(t))$$

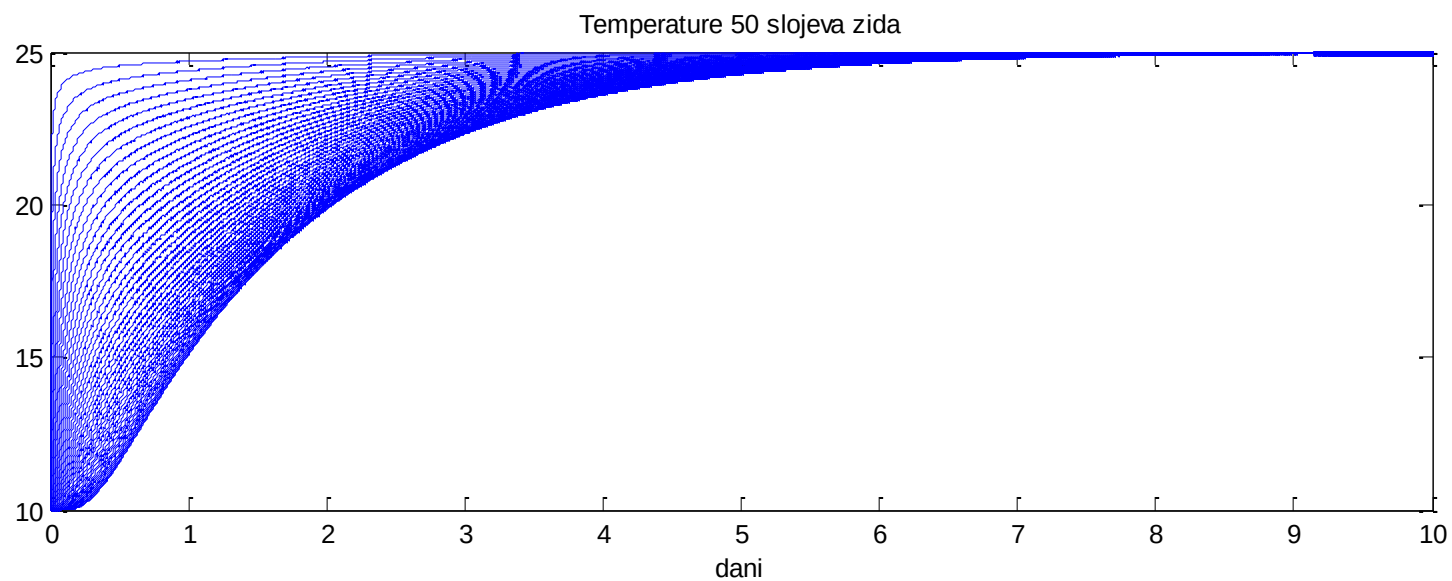
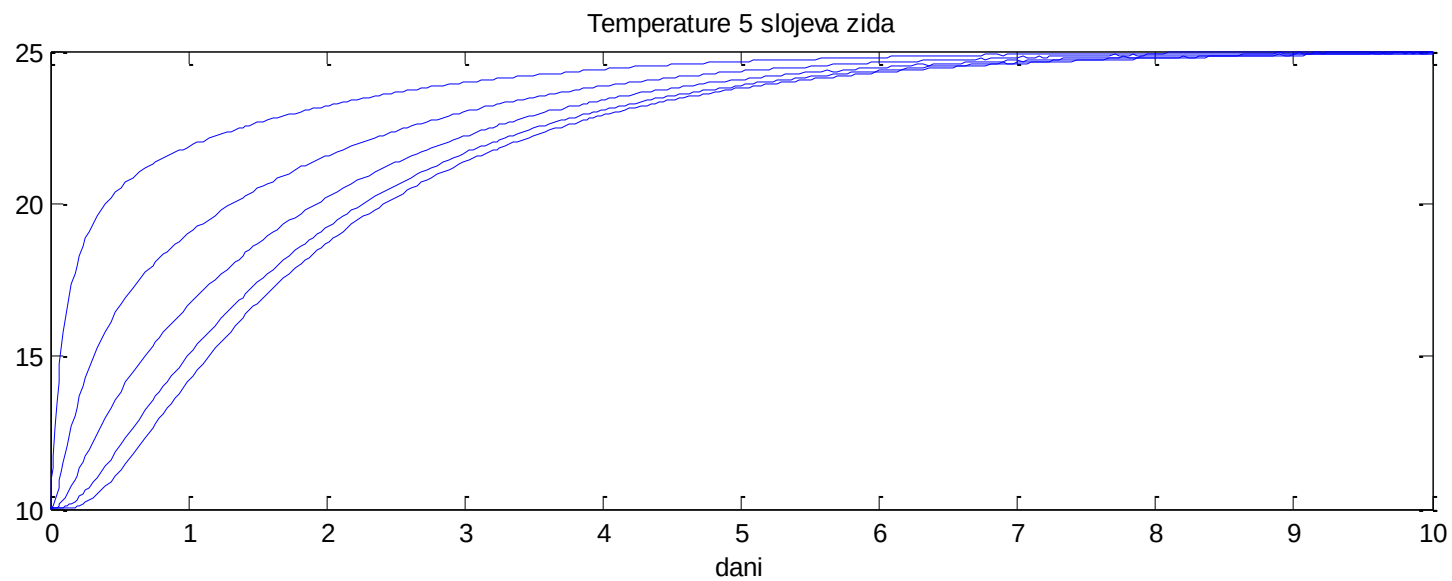
Promena temperature u zidu - program

```
ro = 2400;           % gustina betona [kg/m3]
cp = 800;            % spec. topl. tela [J/(kg.K)]
alfa = 1.5;          % term. prov. [W/(m.K)]
A = 1;               % površina [m2]
L = 0.5;             % debljina zida [m]

R = L/(A*alfa);
C = cp*ro*A*L;
Tpoc = 10;
Tspolj = 25;

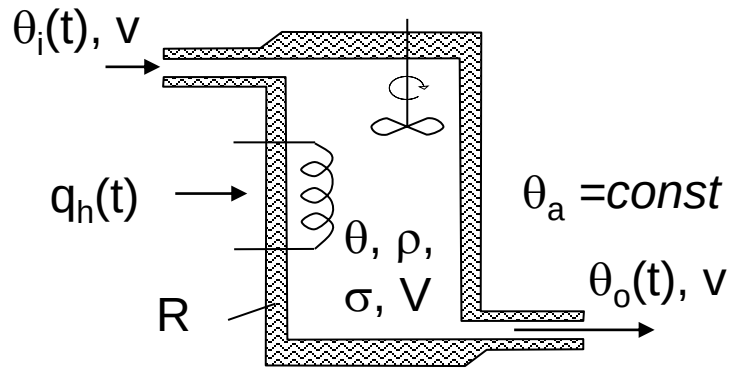
n=5;                 % broj slojeva
[t,T]=ode45(@f,[0:3600:3600*24*10],ones(n,1)*Tpoc,[],n,Tspolj,R,C);
plot(t/86400,T,'b-'), title('Temperature slojeva zida'), xlabel('dani')

function Tprim = f(t,T,n,Ts,R,C)
Tprim = ([Ts;T(1:end-1)] - 2*T + [T(2:end);T(end)])*n*n/R/C;
```



Primer

- Formirati model sistema



$$C = \sigma m = \sigma \rho V$$

$$q_{in} = q_h(t) + v\rho\sigma\theta_i(t)$$

$$q_{out} = v\rho\sigma\theta(t) + \frac{1}{R}(\theta - \theta_a)$$

Napomene:

- Voda u rezervoaru se meša tako da je temperatura θ u njemu homogena.
- Zapreminski protok $v = \text{const}$
- $\theta_o = \theta$ zbog mešanja

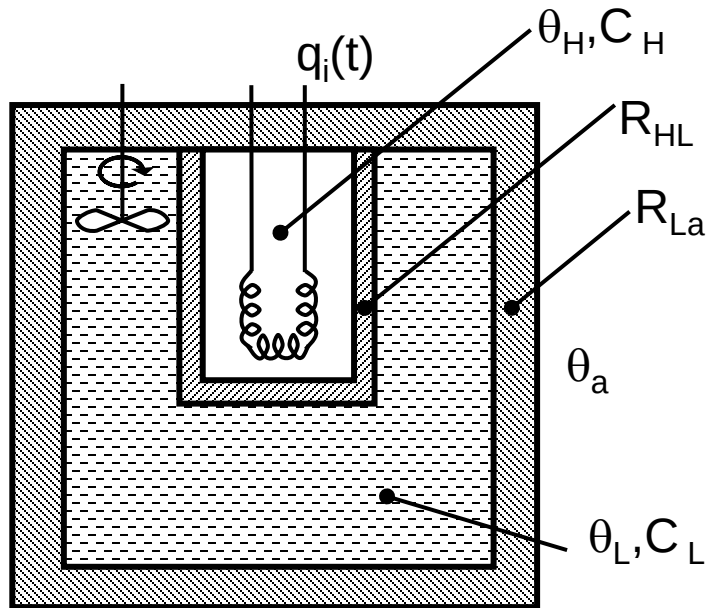
$$\dot{\theta}(t) = \frac{1}{C}(q_{in} - q_{out})$$

$$\dot{\theta} + \left(\frac{v}{V} + \frac{1}{RC} \right) \theta = \frac{v}{V} \theta_i(t) + \frac{1}{C} q_h(t) + \frac{1}{RC} \theta_a$$

$$\frac{1}{\tau} = \frac{v}{V} + \frac{1}{RC}$$

$$\Delta q = \frac{d}{dt}(C \cdot \theta) = \theta \frac{dC}{dt} = \theta \sigma \rho \frac{dV}{dt} = \theta \sigma \rho v$$

Primer



$$C_H = 20000$$

$$C_L = 10^6$$

$$R_{HL} = 10^{-3}$$

$$R_{La} = 5 \cdot 10^{-3}$$

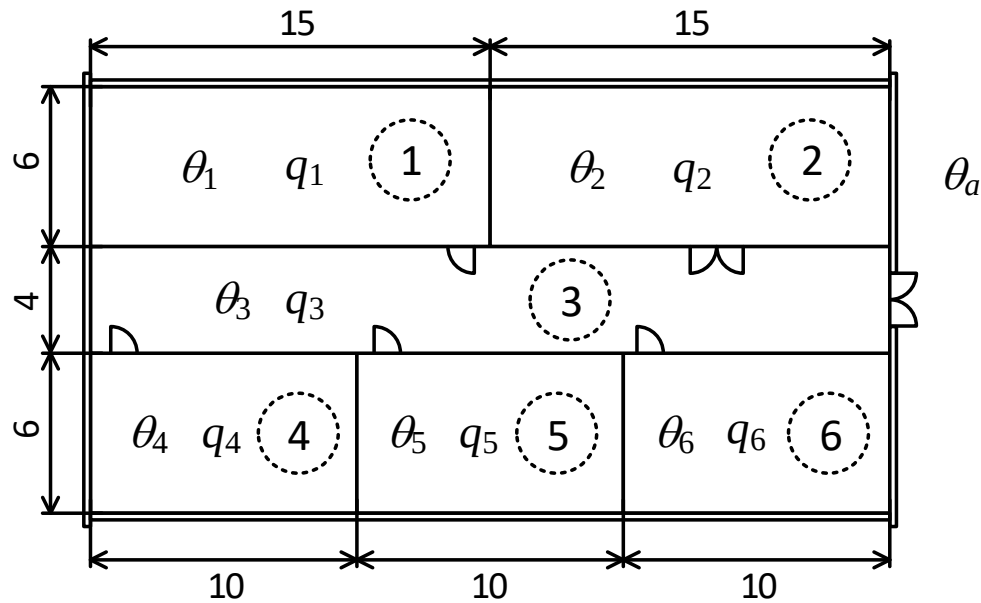
$$\theta_a = 300\text{K}$$

$$\dot{\theta}_H = \frac{1}{C_H} \left(q_{in}(t) - \frac{1}{R_{HL}} (\theta_L - \theta_a) \right)$$

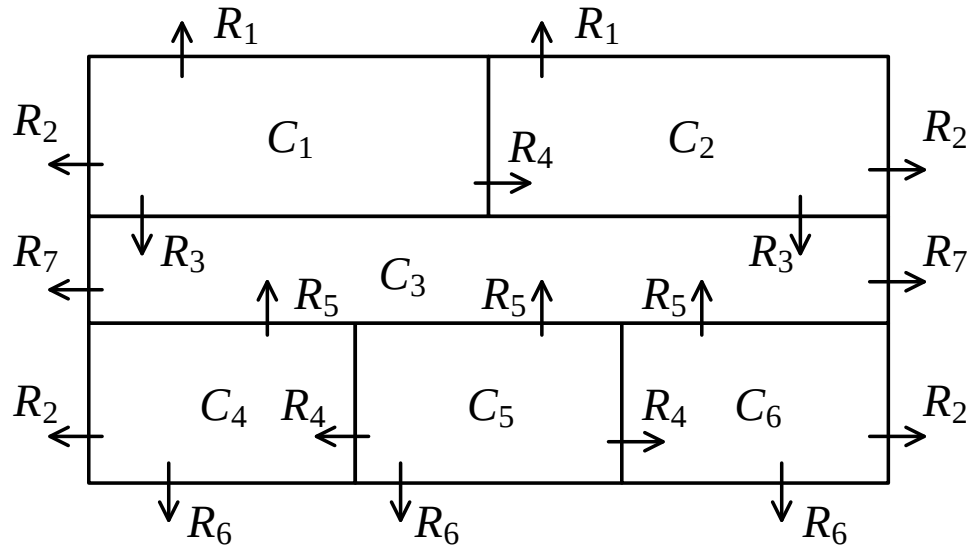
$$\dot{\theta}_L = \frac{1}{C_L} \left(\frac{1}{R_{HL}} (\theta_H - \theta_L) - \frac{1}{R_{La}} (\theta_L - \theta_a) \right)$$

Ako je $q_i(t) = 1,5 \cdot 10^4 \cdot h(t)$ odrediti vreme potrebno da se θ_L zagreje do 365K

Primer: Grejanje zgrade



Formiranje modela



$$\dot{\theta}_1 = \frac{1}{C_1} \left(q_1 - \left(\frac{1}{R_1} + \frac{1}{R_2} \right) (\theta_1 - \theta_a) - \frac{1}{R_3} (\theta_1 - \theta_3) - \frac{1}{R_4} (\theta_1 - \theta_2) \right)$$

$$\dot{\theta}_1 = \frac{1}{C_1} \left(q_1 - \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right) \theta_1 + \frac{1}{R_4} \theta_2 + \frac{1}{R_3} \theta_3 + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \theta_a \right)$$

Dinamički model

$$\dot{\theta}_1 = \frac{1}{C_1} \left(q_1 - \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right) \theta_1 + \frac{1}{R_4} \theta_2 + \frac{1}{R_3} \theta_3 + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \theta_a \right)$$

$$\dot{\theta}_2 = \frac{1}{C_2} \left(q_2 + \frac{1}{R_4} \theta_1 - \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right) \theta_2 + \frac{1}{R_3} \theta_3 + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \theta_a \right)$$

$$\dot{\theta}_3 = \frac{1}{C_3} \left(q_3 + \frac{1}{R_3} \theta_1 + \frac{1}{R_3} \theta_2 - \left(\frac{2}{R_3} + \frac{3}{R_5} + \frac{2}{R_7} \right) \theta_3 + \frac{1}{R_5} \theta_4 + \frac{1}{R_5} \theta_4 + \frac{1}{R_5} \theta_6 + \frac{2}{R_7} \theta_a \right)$$

$$\dot{\theta}_4 = \frac{1}{C_4} \left(q_4 + \frac{1}{R_5} \theta_3 - \left(\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_5} + \frac{1}{R_6} \right) \theta_4 + \frac{1}{R_4} \theta_5 + \left(\frac{1}{R_2} + \frac{1}{R_6} \right) \theta_a \right)$$

$$\dot{\theta}_5 = \frac{1}{C_5} \left(q_5 + \frac{1}{R_5} \theta_3 + \frac{1}{R_4} \theta_4 - \left(\frac{2}{R_4} + \frac{1}{R_5} + \frac{1}{R_6} \right) \theta_5 + \frac{1}{R_4} \theta_6 + \frac{1}{R_6} \theta_a \right)$$

$$\dot{\theta}_6 = \frac{1}{C_6} \left(q_6 + \frac{1}{R_5} \theta_3 + \frac{1}{R_4} \theta_5 - \left(\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_5} + \frac{1}{R_6} \right) \theta_6 + \left(\frac{1}{R_2} + \frac{1}{R_6} \right) \theta_a \right)$$

Statički model

$$\bar{q}_1 - \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right) \bar{\theta}_1 + \frac{1}{R_4} \bar{\theta}_2 + \frac{1}{R_3} \bar{\theta}_3 + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \theta_a = 0$$

$$\bar{q}_2 + \frac{1}{R_4} \bar{\theta}_1 - \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} \right) \bar{\theta}_2 + \frac{1}{R_3} \bar{\theta}_3 + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \theta_a = 0$$

$$\bar{q}_3 + \frac{1}{R_3} \bar{\theta}_1 + \frac{1}{R_3} \bar{\theta}_2 - \left(\frac{2}{R_3} + \frac{3}{R_5} + \frac{2}{R_7} \right) \bar{\theta}_3 + \frac{1}{R_5} \bar{\theta}_4 + \frac{1}{R_5} \bar{\theta}_5 + \frac{1}{R_5} \bar{\theta}_6 + \frac{2}{R_7} \theta_a = 0$$

$$\bar{q}_4 + \frac{1}{R_5} \bar{\theta}_3 - \left(\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_5} + \frac{1}{R_6} \right) \bar{\theta}_4 + \frac{1}{R_4} \bar{\theta}_5 + \left(\frac{1}{R_2} + \frac{1}{R_6} \right) \theta_a = 0$$

$$\bar{q}_5 + \frac{1}{R_5} \bar{\theta}_3 + \frac{1}{R_4} \bar{\theta}_4 - \left(\frac{2}{R_4} + \frac{1}{R_5} + \frac{1}{R_6} \right) \bar{\theta}_5 + \frac{1}{R_4} \bar{\theta}_6 + \frac{1}{R_6} \theta_a = 0$$

$$\bar{q}_6 + \frac{1}{R_5} \bar{\theta}_3 + \frac{1}{R_4} \bar{\theta}_5 - \left(\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_5} + \frac{1}{R_6} \right) \bar{\theta}_6 + \left(\frac{1}{R_2} + \frac{1}{R_6} \right) \theta_a = 0$$

Vektorski zapis ...

$$\underline{A\bar{\theta} + \bar{Q} + b\theta_a = 0}$$

$$\bar{\theta} = \begin{bmatrix} \bar{\theta}_1 \\ \bar{\theta}_2 \\ \bar{\theta}_3 \\ \bar{\theta}_4 \\ \bar{\theta}_5 \\ \bar{\theta}_6 \end{bmatrix}$$

$$\bar{Q} = \begin{bmatrix} \bar{q}_1 \\ \bar{q}_2 \\ \bar{q}_3 \\ \bar{q}_4 \\ \bar{q}_5 \\ \bar{q}_6 \end{bmatrix}$$

$$A = \begin{bmatrix} a_1 & \frac{1}{R_4} & \frac{1}{R_3} & 0 & 0 & 0 \\ \frac{1}{R_4} & a_2 & \frac{1}{R_3} & 0 & 0 & 0 \\ \frac{1}{R_3} & \frac{1}{R_3} & a_3 & \frac{1}{R_5} & \frac{1}{R_5} & \frac{1}{R_5} \\ 0 & 0 & \frac{1}{R_5} & a_4 & \frac{1}{R_4} & 0 \\ 0 & 0 & \frac{1}{R_5} & \frac{1}{R_4} & a_5 & \frac{1}{R_4} \\ 0 & 0 & \frac{1}{R_5} & 0 & \frac{1}{R_4} & a_6 \end{bmatrix}$$

$$a_1 = -\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}\right)$$

$$a_2 = -\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}\right)$$

$$a_3 = -\left(\frac{2}{R_3} + \frac{3}{R_5} + \frac{2}{R_7}\right)$$

$$a_4 = -\left(\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_5} + \frac{1}{R_6}\right)$$

$$a_5 = -\left(\frac{2}{R_4} + \frac{1}{R_5} + \frac{1}{R_6}\right)$$

$$a_6 = -\left(\frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_5} + \frac{1}{R_6}\right)$$

$$b = \begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} \\ \frac{1}{R_1} + \frac{1}{R_2} \\ \frac{2}{R_7} \\ \frac{1}{R_2} + \frac{1}{R_6} \\ \frac{1}{R_6} \\ \frac{1}{R_2} + \frac{1}{R_6} \end{bmatrix}$$